

Synthetic Aromatic Amino Acids from a Negishi Cross-Coupling Reaction

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Received 26 June 2009; revised 28 September 2009

Abstract: An N,O-protected, iodinated bishomoalanine derivative, safely available from glutamic acid, reacts with aryl halides in a Negishi reaction in high yields. From the coupling product, Fmoc-protected amino acids with aromatic and heteroaromatic side chains were generated in high yields by racemization-free procedures. These monomers could be used for solid-phase peptide synthesis.

Key words: amino alcohols, chiral pool, nickel catalyst, RNA ligand, zinc organyls

RNA with its various secondary and tertiary structures offers binding sites for antibiotics, proteins, and small molecules.¹ It can, thus, be considered an interesting drug target. Recently, we reported on basic tripeptides that bind to the TAR element of HIV-1.² The association with TAR blocks the Tat/TAR complex, an important molecular switch regulating transcription efficacy in HIV,³ and thereby inhibits the replication of HIV in cell culture experiments.²

Complex formation between RNA and ligands is governed by hydrogen bonds, Coulomb forces, and stacking interactions. If peptide ligands are considered, the main contribution to stacking comes from aromatic amino acids and thus is limited to His, Phe, Trp, and Tyr. Our concept to expand the potency of peptidic RNA ligands was to start from a broader range of building blocks capable of stacking with nucleobases. Towards this aim we are interested in the preparation of artificial amino acids with aromatic moieties.

For the synthesis of non-natural amino acids various methods exist.^{4,5} Some convert prochiral starting materials enantioselectively into the amino acid precursor, which is most elegant in a formal sense. This approach requires, on the other hand, the optical purity to be verified for each new amino acid and, if necessary, the level of enantiomeric excess to be enriched in a further purification step.

Due to this drawback, we have adopted a different strategy based on racemization proof transformations of enantiomerically pure common building blocks. In previous studies, derivatives of allylglycine⁶ and vinylglycinol^{6,7} have been used as starting materials that were further functionalized to aromatic amino acids by Suzuki or Heck

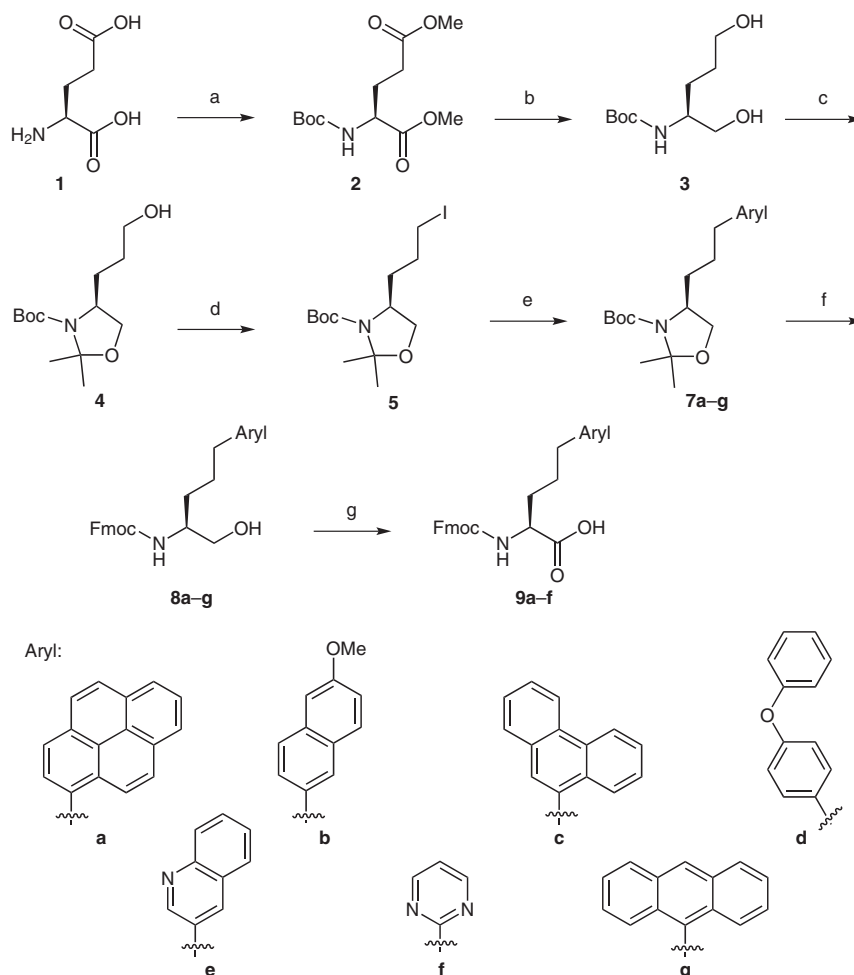
cross-coupling reactions. Here we report on the conversion of glutamic acid into the enantiomerically pure bis-homoalaninol derivative **5**. Compound **5** serves as a key intermediate that, after halogen–zinc exchange, can be transformed into a broad range of amino acid precursors by Negishi cross coupling with aryl and heteroaryl halides. Similar Negishi reactions⁸ of iodoalanine and other iodinated amino acid derivatives have been previously achieved by Jackson et al.⁹

After the esterification of L-glutamic acid (**1**) with chlorotrimethylsilane in methanol and Boc protection in quantitative yield, both ester groups were reduced under mild conditions with lithium borohydride (Scheme 1). Lower yields were obtained in this step using lithium aluminum hydride. Almost no product was isolated with sodium borohydride, even in the presence of lithium chloride. Selective N,O-ketalization of aminodiol **3** with 2,2-dimethoxypropane then formed the oxazolidine moiety of **4**,¹⁰ which is thermodynamically favored over acetonides with medium-sized rings. To obtain building block **5**, the remaining primary alcohol was converted into the alkyl halide with iodine/triphenylphosphine.

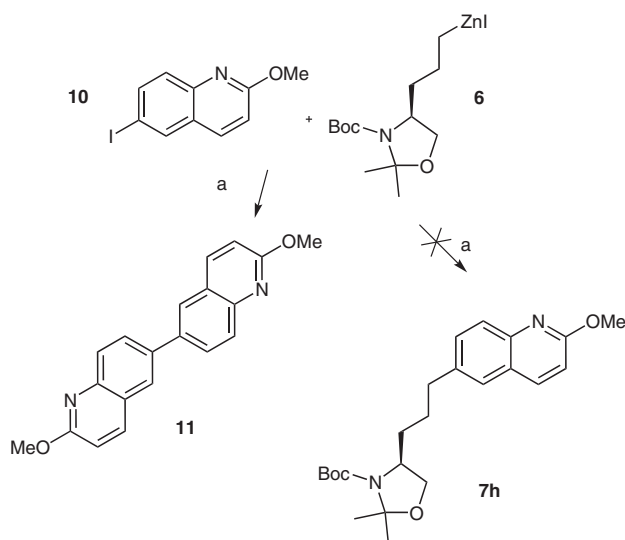
Zinc organyl **6** (Scheme 2) was prepared from **5** with iodine-activated zinc powder.^{11,12} The cross-coupling step of reagent **6** and aryl bromides proved troublesome at the beginning when palladium catalysts were used. The key to success was to apply NiCl₂(PPh₃)₂ instead. Finally, variation of the solvent, reaction time, and temperature led to the optimal conditions (DMF, 80 °C, 3 h) producing compounds **7a–g** at gram scale.¹² The use of aryl iodides, which was tested exemplarily for 2-iodo-6-methoxynaphthalene, did not increase the yield significantly.

The quinoline derivative **10** represents an exception. Here, the desired Negishi product **7h** was not formed at all. Instead, almost quantitative homocoupling of the aromatic system was observed (Scheme 2).

With cross-coupling products **7** made available, just a few steps were required to obtain the corresponding α -amino acids **9**. At first, the protecting groups were removed simultaneously under acidic conditions. The amino residue was then reprotected with *N*-{[(9-fluorenylmethoxy)carbonyl]oxy}succinimide (Fmoc-OSu) to give **8**. Oxidation of the hydroxy group with pyridinium dichromate in *N,N*-dimethylformamide completed the synthesis of amino acids **9**.¹³ For some Fmoc-amino alcohols a slight variation with catalytic amounts of pyridinium dichromate and periodic acid as principal oxidant proved superior.¹⁴ In a sin-

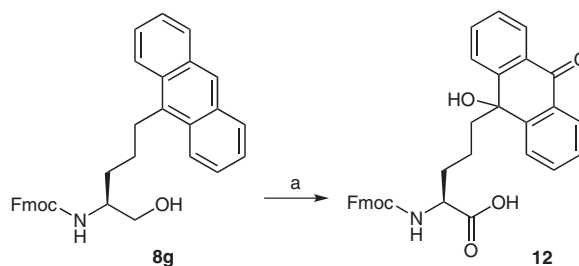


Scheme 1 Reagents and conditions: (a) 1. TMSCl, MeOH, 2. Boc₂O, Et₃N, quant. (2 steps); (b) LiBH₄, THF, 93%; (c) Me₂C(OMe)₂, PTSA, 91%; (d) I₂, Ph₃P, DIPEA, 91%; (e) 1. Zn, I₂, 2. Ar-X, NiCl₂(PPh₃)₂, 80–95%; (f) 1. TFA, 2. Fmoc-OSu, 75–94%, (g) PDC, DMF, 79–92%, >99% ee.



Scheme 2 Homocoupling under Negishi conditions. Reagents and conditions: (a) NiCl₂(PPh₃)₂, DMF, 80 °C, 3 h.

gle case, amino alcohol **8g**, this final step failed due to oxidative degradation of anthracene forming the hydroxy-



Scheme 3 Oxidation of the anthracene amino alcohol **8g** with PDC in DMF

anthrone **12** (Scheme 3).⁷ The optimized reaction conditions and yields of all steps are shown below (Table 1).

To rule out racemization in our new synthetic route, we prepared *ent*-**9a** from D-glutamic acid (*ent*-**1**). The availability of the two enantiomers **9a** and *ent*-**9a** allowed us to verify >99% ee by chiral HPLC.

Amino acids **9** are characterized by a C₃-linker between the glycine framework and the aromatic moiety. Applying the methods shown in Scheme 1, we also succeeded in synthesizing pyrene-substituted amino alcohol **19** with a C₂-linker from L-aspartic acid (Scheme 4). Compound **19**

Table 1 Reaction Conditions and Yields

7–9	Yield (%) of 7	Yield (%) of 8	Yield (%) of 9
a	88 ^a	91 ^c	83 ^c
b	84 ^b	78 ^c	80 ^c
c	85 ^a	88 ^c	92 ^c
d	95 ^a	75 ^c	70 ^f
e	87 ^a	85 ^c	82 ^f
f	86 ^a	94 ^d	79 ^g
g	80 ^a	85 ^c	–

^a (1) Zn, I₂, DMF, 40 °C, 1 h; (2) NiCl₂(PPh₃)₂, ArBr, 80 °C, 3 h.^b (1) Zn, I₂, DMF, 40 °C, 1 h; (2) NiCl₂(PPh₃)₂, 2-iodo-6-methoxynaphthalene, 80 °C, 3 h.^c (1) TFA, r.t., 1 h; (2) Fmoc-OSu, EtOAc–EtOH (2:1), r.t., ~2 h.^d (1) TFA, r.t., 1 h; (2) Fmoc-OSu, DIPEA, EtOAc–EtOH (2:1), r.t., ~2 h.^e PDC, DMF, 40 °C, 8 h.^f PDC, H₅IO₆, DMF, 40 °C, 8 h.^g PDC, H₅IO₆, acetone, r.t., overnight.

has been prepared previously from vinylglycinol using the Heck pathway. Its conversion into amino acid **20** is known.⁷

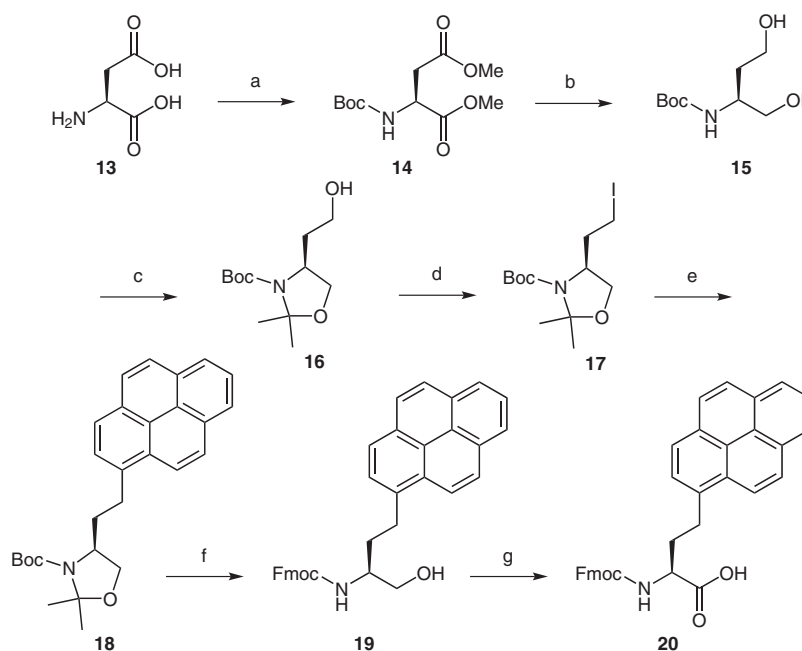
In summary, we have developed an efficient synthesis of Fmoc-protected aromatic α -amino acids from L- or D-glutamic acid. The enantiopure building block **5** was converted easily into the organozinc species **6** which undergoes nickel-catalyzed Negishi cross coupling in high yields (80–95%). Our synthesis comprises ten steps in total with combined yields up to 53%. The final building blocks **9** are ready for further application in solid-phase

peptide synthesis. In our previous work, we have synthesized amino acids of type **9** (C₃-linker) from an allylglycinol intermediate obtained in a diastereoselective Myers alkylation.⁶ This approach is more laborious on a large scale compared to the preparation of **5**. Methionine, on the other hand, can be easily transformed into enantiomerically pure vinylglycinol derivatives.⁶ Using either Suzuki⁶ or Heck⁷ cross-coupling reactions, we have prepared C₂-linker amino acids such as **20** from this building block. Apart from the linker length (C₂ is also available from aspartic acid, Scheme 4), how do the methods compare? Compound **5** and the vinylglycinol derivative are both accessible in enantiomerically pure form in 30-gram batches. The yields of the cross-coupling, deprotection, and oxidation steps of both routes are almost equivalent. However, a major practical advantage of the new method is the fast and simplified deprotection of intermediates **7** and **18** (step f in Schemes 1 and 4) reducing the time requirement of the synthesis. The preparation and evaluation of peptides prepared from building blocks **9** and **20** will be reported elsewhere.

NMR: Bruker DPX 250 (¹H: 250 MHz; ¹³C: 62.9 MHz), Bruker AM 300 (¹H: 300 MHz; ¹³C: 75.4 MHz) or Bruker Avance 400 (¹H: 400 MHz; ¹³C: 100.6 MHz). FT-IR: Perkin-Elmer 1600 Series or Jasco 420. Elementary analysis: Heraeus CHN Rapid. Mass spectrometry: Fisons VG Platform II (ESI); Fisons VG Tofspec (MALDI). Melting points (uncorrected): Kofler hot plate microscope. Optical rotation: Perkin-Elmer polarimeter 241.

Dimethyl (2S)-2-[(*tert*-Butoxycarbonyl)amino]pentanedioate (**2**)

To a 1000-mL round-bottom flask containing ice-cold MeOH (300 mL) was added TMSCl (108 mL, 850 mmol) and then L-glutamic acid (**1**, 25.0 g, 170 mmol) was added in small portions. The mixture



Scheme 4 Reagents and conditions: (a) (1) TMSCl, MeOH, (2) Boc₂O, Et₃N, 95% (2 steps); (b) LiBH₄, THF, 88%; (c) Me₂C(OMe)₂, PTSA, 85%; (d) I₂, Ph₃P, DIPEA, 49%; (e) 1. Zn, I₂, 2. 1-bromopyrene, NiCl₂(PPh₃)₂, 64%; (f) 1. TFA, 2. Fmoc-OSu, 68%; (g) see ref. 7. Yields were not optimized.

was stirred at r.t. overnight. After quantitative conversion, the mixture was cooled to 0 °C. Et₃N (153 mL, 1.1 mol) and Boc₂O (40.8 g, 187 mmol) were added slowly to the soln. The turbid mixture was diluted with MeOH (100 mL) and stirred again at r.t. overnight. The mixture was concentrated in vacuo, the residue was suspended in Et₂O (500 mL) and filtered through Celite. The filtrate was concentrated under reduced pressure and purified by column chromatography (*n*-hexane–EtOAc, 2:1) to give **2** (46.8 g, quant.) as a colorless oil; *R*_f = 0.4 (*n*-hexane–EtOAc, 1:1).

[α]_D²⁰ –26.0 (*c* 1.0, MeOH).

IR (film): 3372 (m), 2978 (s), 1714 (s), 1518 (s), 1439 (s), 1367 (s), 1252 (s), 1168 (s), 1051 (m), 1028 (m), 881 (w), 857 (w), 780 (w), 586 cm^{–1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 7.28 (d, *J* = 7.8 Hz, 1 H, NH, D₂O exch.), 3.99 (m, 1 H, CH), 3.62 (s, 3 H, CHCO₂CH₃), 3.58 (s, 3 H, CH₂CO₂CH₃), 2.37 (m, 2 H, CH₂CO₂CH₃), 1.99 (m, 1 H, CHCH_a), 1.77 (m, 1 H, CHCH_b), 1.37 [s, 9 H, C(CH₃)₃].

¹³C NMR (62.9 MHz, DMSO-*d*₆): δ = 173.6, 172.5, 155.4, 78.2, 52.5, 51.7, 51.3, 29.6, 28.1, 25.8.

Anal. Calcd for C₁₂H₂₁NO₆: C, 52.35; H, 7.69; N, 5.09. Found: C, 52.33; H, 7.77; N, 5.24.

Dimethyl (2*R*)-2-[(*tert*-Butoxycarbonyl)amino]pentanedioate (*ent*-2)

Compound *ent*-2 was synthesized by the same procedure. NMR and IR spectra are identical with those of **2**.

[α]_D²⁰ +26.0 (*c* 1.0, MeOH).

Anal. Calcd for C₁₂H₂₁NO₆: C, 52.35; H, 7.69; N, 5.09. Found: C, 52.47; H, 7.87; N, 5.10.

tert-Butyl *N*-[(2*S*)-1,5-Dihydroxypentan-2-yl]carbamate (**3**)

In a flame-dried Schlenk vessel, THF (150 mL) was cooled in an ice bath. LiBH₄ (5.00 g, 230 mmol) was added under argon. Then **2** (12.5 g, 45.9 mmol) in THF (30 mL) was added dropwise to the soln. The mixture was stirred at 0 °C for 1 h, then at r.t. overnight. MeOH was added dropwise until gas evolution ceased. The turbid soln was concentrated under reduced pressure. The residue was taken up with H₂O and extracted with EtOAc (3 × 120 mL). The combined organic phases were dried (MgSO₄) and concentrated in vacuo. The residue was purified by column chromatography (EtOAc–*n*-hexane, 10:1). The product was dried in vacuo to give **3** (9.34 g, 93%) as a colorless crystalline solid; mp 64–65 °C; *R*_f = 0.2 (EtOAc–*n*-hexane, 10:1).

[α]_D²⁰ –15.1 (*c* 1.0, MeOH).

IR (KBr): 3347 (s), 2987 (m), 2941 (m), 2919 (s), 2867 (m), 1682 (s), 1534 (s), 1450 (m), 1389 (w), 1368 (m), 1300 (m), 1254 (m), 1171 (m), 1122 (w), 1075 (m), 1006 (m), 990 (m), 903 (w), 871 (w), 781 (w), 749 cm^{–1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 6.43 (d, *J* = 8.3 Hz, 1 H, NH, D₂O exch.), 4.53 (t, *J* = 5.3 Hz, 1 H, CHCH₂OH, D₂O exch.), 4.34 (t, *J* = 5.3 Hz, 1 H, CH₂CH₂OH, D₂O exch.), 3.39–3.15 (m, 5 H, CH, CH₂OH), 1.52 (m, 2 H, CH₂), 1.37 [s, 9 H, C(CH₃)₃], 1.22 (m, 2 H, CH₂).

¹³C NMR (62.9 MHz, DMSO-*d*₆): δ = 155.3, 77.2, 63.6, 60.8, 52.1, 29.1, 28.2, 27.5.

Anal. Calcd for C₁₀H₂₁NO₄: C, 54.77; H, 9.65; N, 6.39. Found: C, 54.96; H, 9.83; N, 6.36.

tert-Butyl *N*-[(2*R*)-1,5-Dihydroxypentan-2-yl]carbamate (*ent*-3)

Compound *ent*-3 was synthesized by the same procedure. NMR and IR spectra are identical with those of **3**.

[α]_D²⁰ +15.1 (*c* 1.0, MeOH).

Anal. Calcd for C₁₀H₂₁NO₄: C, 54.77; H, 9.65; N, 6.39. Found: C, 54.94; H, 9.84; N, 6.37.

tert-Butyl (4*S*)-4-(3-Hydroxypropyl)-2,2-dimethyloxazolidine-3-carboxylate (**4**)

Diol **3** (29.0 g, 132 mmol) was dissolved in CH₂Cl₂ (300 mL). 2,2-Dimethoxypropane (243 mL, 1.98 mol) and PTSA·H₂O (2.51 g, 13.2 mmol) were added. The mixture was stirred at r.t. for exactly 90 min. Sat. NaHCO₃ (150 mL) was added and the aqueous phase was extracted with EtOAc (3 × 150 mL). The combined organic phases were dried (MgSO₄), concentrated in vacuo, and purified by column chromatography (*n*-hexane–EtOAc, 10:1) to give **4** (31.3 g, 91%) as a colorless oil; *R*_f = 0.6 (*n*-hexane–EtOAc, 2:1).

[α]_D²⁰ +36.4 (*c* 0.5, MeOH).

IR (film): 3447 (s), 2979 (s), 2936 (s), 2871 (m), 1698 (s), 1478 (m), 1457 (m), 1390 (s), 1257 (m), 1207 (m), 1175 (s), 1148 (m), 1087 (s), 1030 (m), 946 (w), 845 (m), 806 (m), 769 (m), 662 (w), 560 cm^{–1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 4.39 (t, *J* = 5.3 Hz, 1 H, OH, D₂O exch.), 3.88 (dd, *J* = 8.5 Hz, *J* = 5.5 Hz, 1 H, CHCHHO), 3.74 (m, 1 H, NCH), 3.68 (dd, *J* = 8.5 Hz, *J* = 1.25 Hz, 1 H, CHCHHO), 3.38 (dt, *J* = 6.3 Hz, *J* = 5.3 Hz, 2 H, CH₂OH), 1.64 (m, 2 H, CH₂), 1.47 (s, 3 H, CH₃), 1.41 (s, 9 H, CH₂, CH₃), 1.39–1.32 (m, 5 H, CH₃).¹⁵

¹³C NMR (62.9 MHz, DMSO-*d*₆): δ = 151.0, 92.4, 78.8, 78.4, 66.4, 66.0, 60.5, 56.7, 29.9, 29.0, 27.9, 27.3, 26.5, 24.3, 22.9.¹⁵

Anal. Calcd for C₁₃H₂₅NO₄: C, 60.21; H, 9.72; N, 5.40. Found: C, 60.00; H, 9.91; N, 5.43.

tert-Butyl (4*R*)-4-(3-Hydroxypropyl)-2,2-dimethyloxazolidine-3-carboxylate (*ent*-4)

Compound *ent*-4 was synthesized by the same procedure. NMR and IR spectra are identical with those of **4**.

[α]_D²⁰ –36.4 (*c* 0.5, MeOH).

Anal. Calcd for C₁₃H₂₅NO₄: C, 60.21; H, 9.72; N, 5.40. Found: C, 60.30; H, 9.46; N, 5.57.

tert-Butyl (4*S*)-4-(3-Iodopropyl)-2,2-dimethyloxazolidine-3-carboxylate (**5**)

Ph₃P (27.6 g, 105 mmol) was dissolved in CH₂Cl₂ (250 mL) and I₂ (26.7 g, 105 mmol) was added to the clear soln. After 5 min, DIPEA (18.0 mL, 105 mmol) and alcohol **4** [24.8 g, 95.7 mmol, dissolved in CH₂Cl₂ (50 mL)] were added. The mixture was stirred at r.t. for 2 h. Et₂O (300 mL) was added and the precipitate was filtered through Celite. The filtrate was washed successively with sat. NaHCO₃, Na₂S₂O₅ soln, and H₂O. The soln was dried (MgSO₄) and the solvent evaporated to give an oily brown residue that was purified by column chromatography (*n*-hexane–EtOAc, 10:1). The product fractions were concentrated in vacuo and stored at –25 °C to give **5** (32.2 g, 91%) as a colorless solid; mp 32 °C; *R*_f = 0.75 (*n*-hexane–EtOAc, 3:1).

[α]_D²⁰ +26.8 (*c* 1.0, MeOH).

IR (KBr): 2978 (m), 2936 (m), 2874 (w), 1693 (s), 1478 (m), 1455 (m), 1390 (s), 1313 (m), 1256 (s), 1208 (m), 1174 (s), 1096 (s), 977 (w), 860 (m), 846 (m), 806 (w), 769 (m), 558 cm^{–1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 3.88 (dd, *J* = 8.5 Hz, *J* = 5.5 Hz, 1 H, CHCHHO), 3.76 (m, 1 H, NCH), 3.69 (d, *J* = 8.5 Hz, 1 H, CHCHHO), 3.29 (m, 2 H, CH₂I), 1.79–1.54 [m, 4 H, CH(CH₂)₂], 1.47 (s, 3 H, CH₃), 1.42 (s, 9 H, CH₃), 1.38 (s, 3 H, CH₃).¹⁵

¹³C NMR (62.9 MHz, DMSO-*d*₆): δ = 150.9, 92.6, 78.8, 78.6, 66.6, 66.2, 55.8, 34.1, 33.6, 29.7, 27.9, 27.3, 26.4, 24.2, 22.8, 8.6, 8.3.¹⁵

Anal. Calcd for C₁₃H₂₄INO₃: C, 42.29; H, 6.55; N, 3.79. Found: C, 42.46; H, 6.64; N, 3.84.

***tert*-Butyl (4*R*)-4-(3-Iodopropyl)-2,2-dimethyloxazolidine-3-carboxylate (*ent*-5)**

Compound *ent*-5 was synthesized by the same procedure. NMR and IR spectra are identical with those of 5.

$[\alpha]_{\text{D}}^{20}$ –26.8 (*c* 1.0, MeOH).

Anal. Calcd for $\text{C}_{13}\text{H}_{24}\text{INO}_3$: C, 42.29; H, 6.55; N, 3.79. Found: C, 42.36; H, 6.57; N, 3.91.

***tert*-Butyl 4-(3-Arylpropyl)-2,2-dimethyloxazolidine-3-carboxylates 7a–g; General Procedure 1**

A 2-necked flask charged with Zn powder (0.72 g, 11.1 mmol) and I_2 (tip of a spatula) was flame dried for 10 min in vacuo. After flushing with argon, anhyd DMF (7 mL) and alkyl iodide 5 (3.00 g, 8.13 mmol) were added and the mixture was stirred at 40 °C for 1 h (TLC monitoring; the reaction was regarded as finished when only a baseline spot was visible). After cooling to r.t., aryl halide (7.39 mmol) and $\text{NiCl}_2(\text{PPh}_3)_2$ (0.48 g, 0.74 mmol) were added under argon. The mixture was stirred at 80 °C for 3 h. After cooling, brine (30 mL) and EtOAc (40 mL) were added and the mixture was filtered through Celite. The aqueous phase was extracted with EtOAc (2 × 100 mL). The combined organic phases were dried (MgSO_4) and concentrated in vacuo. The residue was purified by column chromatography.

***tert*-Butyl (4*S*)-2,2-Dimethyl-4-[3-(pyren-1-yl)propyl]oxazolidine-3-carboxylate (7a)**

Following general procedure 1 using 1-bromopyrene (2.27 g) and column chromatography (*n*-hexane–EtOAc, 25:1 then 10:1) gave 7a as a highly viscous oil; yield: 2.88 g (88%); >99% ee [HPLC analysis (Chiral Whelk 01 RR column, *n*-hexane–*i*-PrOH, 10:1, 1.0 mL/min, 254 nm): t_{R} = 6.90 (*R*), 10.85 min (*S*); R_{f} = 0.5 (*n*-hexane–EtOAc, 5:1)].

$[\alpha]_{\text{D}}^{20}$ –10.5 (*c* 0.5, MeOH).

IR (film): 3040 (m), 2977 (m), 2934 (m), 2865 (m), 1694 (s), 1603 (w), 1509 (w), 1477 (m), 1457 (m), 1389 (s), 1364 (s), 1314 (m), 1257 (m), 1206 (m), 1174 (m), 1143 (m), 1097 (m), 1082 cm^{-1} (m).

^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ = 8.32 (d, J = 9.2 Hz, 1 H, H_{aryl}), 8.25 (m, 2 H, H_{aryl}), 8.19 (m, 2 H, H_{aryl}), 8.11 (m, 2 H, H_{aryl}), 8.05 (m, 1 H, H_{aryl}), 7.93 (m, 1 H, H_{aryl}), 3.85 (m, 1 H, NCH), 3.69 (m, 2 H, OCH_2), 3.33 (m, 2 H, ArCH_2), 1.80–1.67 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$), 1.43–1.34 (m, 10 H, CH_3), 1.06 (s, 5 H, CH_3).¹⁵

^1H NMR (250 MHz, $\text{DMSO}-d_6$, 90 °C): δ = 8.36 (m, 9 H, H_{aryl}), 3.92 (dd, J = 8.5 Hz, J = 6 Hz, 1 H, CHCHHO), 3.84 (m, 1 H, NCH), 3.71 (dd, J = 8.5 Hz, J = 1.8 Hz, 1 H, CHCHHO), 3.37 (m, 2 H, ArCH_2), 1.85–1.64 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$), 1.46 (s, 3 H, CH_3), 1.41 (s, 3 H, CH_3), 1.30 [s, 9 H, $\text{C}(\text{CH}_3)_3$].

^{13}C NMR (75.4 MHz, $\text{DMSO}-d_6$): δ = 150.8, 136.5, 130.7, 130.3, 129.1, 127.9, 127.2, 127.0, 126.3, 125.9, 124.70, 124.67, 124.5, 124.1, 124.0, 123.2, 92.4, 78.9, 78.2, 66.4, 66.1, 56.6, 32.8, 32.3, 27.7, 27.5, 26.4, 24.2, 22.8.¹⁵

Anal. Calcd for $\text{C}_{29}\text{H}_{33}\text{NO}_3$: C, 78.52; H, 7.50; N, 3.16. Found: C, 78.68; H, 7.65; N, 3.17.

***tert*-Butyl (4*R*)-2,2-Dimethyl-4-[3-(pyren-1-yl)propyl]oxazolidine-3-carboxylate (*ent*-7a)**

Compound *ent*-7a was synthesized by the same procedure. NMR and IR spectra are identical with those of 7.

$[\alpha]_{\text{D}}^{20}$ +10.5 (*c* 0.5, MeOH).

Anal. Calcd for $\text{C}_{29}\text{H}_{33}\text{NO}_3$: C, 78.52; H, 7.50; N, 3.16. Found: C, 78.65; H, 7.53; N, 3.10.

***tert*-Butyl (4*S*)-4-[3-(6-Methoxynaphthalen-2-yl)propyl]-2,2-dimethyloxazolidine-3-carboxylate (7b)**

Following general procedure 1 using 2-iodo-6-methoxynaphthalene (2.10 g) and column chromatography (*n*-hexane–EtOAc, 10:1) gave 7b as a colorless solid after recrystallization (MeCN); yield: 2.48 g (84%); mp 95–97 °C; >99% ee [HPLC analysis (Chiral Whelk 01 RR column, *n*-hexane–*i*-PrOH, 2:1, 1.3 mL/min, 254 nm): t_{R} = 3.15 (*R*), 4.13 min (*S*); R_{f} = 0.5 (*n*-hexane–EtOAc, 5:1)].

$[\alpha]_{\text{D}}^{20}$ –19.8 (*c* 1.1, MeOH).

IR (KBr): 2979 (m), 2942 (s), 2875 (m), 1703 (s), 1633 (w), 1605 (m), 1481 (w), 1456 (w), 1391 (s), 1371 (s), 1312 (w), 1257 (m), 1240 (m), 1160 (m), 1142 (m), 1096 (m), 1081 (m), 1026 (m), 889 (w), 856 (m), 815 (w), 769 (w), 677 cm^{-1} (w).

^1H NMR (250 MHz, $\text{DMSO}-d_6$): δ = 7.73 (d, J = 8.8 Hz, 2 H, H_{aryl}), 7.60 (s, 1 H, H_{aryl}), 7.32 (d, J = 8.3 Hz, 1 H, H_{aryl}), 7.26 (d, J = 2.5 Hz, 1 H, H_{aryl}), 7.11 (dd, J = 9 Hz, J = 2.5 Hz, 1 H, H_{aryl}), 3.90–3.69 (m, 3 H, NCHCH_2O), 3.85 (s, 3 H, OCH_3), 2.71 (m, 2 H, ArCH_2), 1.69–1.50 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$), 1.44–1.37 (m, 10 H, CH_3), 1.24 (s, 5 H, CH_3).¹⁵

^{13}C NMR (100.6 MHz, $\text{DMSO}-d_6$): δ = 156.6, 150.9, 136.9, 132.6, 128.6, 128.5, 127.6, 126.5, 125.8, 118.3, 105.6, 92.5, 92.4, 78.9, 78.4, 66.4, 56.6, 55.0, 34.8, 32.7, 32.2, 27.8, 27.4, 26.5, 24.3, 22.9.¹⁵

Anal. Calcd for $\text{C}_{24}\text{H}_{33}\text{NO}_4$: C, 72.15; H, 8.33; N, 3.51. Found: C, 71.91; H, 8.42; N, 3.70.

***tert*-Butyl (4*R*)-4-[3-(6-Methoxynaphthalen-2-yl)propyl]-2,2-dimethyloxazolidine-3-carboxylate (*ent*-7b)**

Compound *ent*-7b was synthesized by the same procedure. NMR and IR spectra are identical with those of 7b.

$[\alpha]_{\text{D}}^{20}$ +19.8 (*c* 1.1, MeOH).

Anal. Calcd for $\text{C}_{24}\text{H}_{33}\text{NO}_4$: C, 72.15; H, 8.33; N, 3.51. Found: C, 72.24; H, 8.41; N, 3.62.

***tert*-Butyl (4*S*)-2,2-Dimethyl-4-[3-(phenanthren-9-yl)propyl]oxazolidine-3-carboxylate (7c)**

Following general procedure 1 using 9-bromophenanthrene (2.64 g) and column chromatography (*n*-hexane–EtOAc, 25:1 then 10:1) gave 7c as a colorless solid after recrystallization (MeCN); yield: 2.64 g (85%); mp 102–103 °C; R_{f} = 0.65 (*n*-hexane–EtOAc, 3:1).

$[\alpha]_{\text{D}}^{20}$ +17.2 (*c* 0.5, MeOH).

IR (KBr): 2984 (m), 2949 (m), 2876 (w), 1690 (s), 1625 (w), 1458 (w), 1388 (s), 1364 (s), 1250 (m), 1179 (m), 1084 (s), 1049 (w), 992 (w), 857 (w), 845 (w), 749 (m), 722 (m), 671 cm^{-1} (w).

^1H NMR (250 MHz, $\text{DMSO}-d_6$): δ = 8.87 (m, 1 H, H_{aryl}), 8.78 (m, 1 H, H_{aryl}), 8.13 (m, 1 H, H_{aryl}), 7.90 (m, 1 H, H_{aryl}), 7.72–7.58 (m, 5 H, H_{aryl}), 3.91–3.70 (m, 3 H, NCHCH_2O), 3.11 (m, 2 H, ArCH_2), 1.87–1.60 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$), 1.47–1.36 (m, 10 H, CH_3), 1.16 (s, 5 H, CH_3).¹⁵

^{13}C NMR (75.4 MHz, $\text{DMSO}-d_6$): δ = 150.9, 136.0, 131.2, 130.5, 130.1, 129.0, 127.8, 126.6, 126.3, 126.0, 125.7, 124.2, 123.3, 122.5, 92.5, 78.4, 66.5, 56.6, 33.0, 32.3, 27.7, 26.4, 26.2, 22.9.¹⁵

Anal. Calcd for $\text{C}_{27}\text{H}_{33}\text{NO}_3$: C, 77.29; H, 7.93; N, 3.34. Found: C, 77.27; H, 8.04; N, 3.23.

***tert*-Butyl (4*S*)-2,2-Dimethyl-4-[3-(4-phenoxyphenyl)propyl]oxazolidine-3-carboxylate (7d)**

Following general procedure 1 using 4-bromophenyl phenyl ether (1.84 g, 1.29 mL) and column chromatography (*n*-hexane–EtOAc, 10:1 then 3:1) gave 7d as a colorless oil; yield: 2.88 g (95%); R_{f} = 0.65 (*n*-hexane–EtOAc, 3:1).

$[\alpha]_{\text{D}}^{20}$ +16.9 (*c* 1.2, MeOH).

IR (film): 2979 (m), 2936 (m), 2865 (m), 1693 (s), 1591 (m), 1507 (m), 1490 (s), 1390 (s), 1238 (s), 1170 (m), 1097 (m), 871 (w), 848 (w), 750 (w), 692 cm^{-1} (w).

^1H NMR (250 MHz, $\text{DMSO}-d_6$): δ = 7.36 (m, 2 H, H_{aryl}), 7.20 (m, 2 H, H_{aryl}), 7.10 (m, 1 H, H_{aryl}), 6.94 (m, 4 H, H_{aryl}), 3.87 (m, 1 H, NCH), 3.70 (m, 2 H, CH_2O), 2.57 (m, 2 H, ArCH_2), 1.57 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$), 1.49–1.38 (m, 10 H, CH_3), 1.33 (s, 5 H, CH_3).¹⁵

^{13}C NMR (62.9 MHz, $\text{DMSO}-d_6$): δ = 157.0, 154.3, 150.9, 137.1, 129.8, 129.6, 122.9, 118.6, 118.0, 92.5, 78.9, 78.4, 66.4, 66.0, 56.6, 34.1, 32.6, 27.9, 27.4, 26.5, 24.2, 22.9.¹⁵

Anal. Calcd for $\text{C}_{25}\text{H}_{33}\text{NO}_4$: C, 72.96; H, 8.08; N, 3.40. Found: C, 72.87; H, 7.98; N, 3.19.

tert-Butyl (4R)-2,2-Dimethyl-4-[3-(4-phenoxyphenyl)propyl]oxazolidine-3-carboxylate (ent-7d)

Compound *ent-7d* was synthesized by the same procedure. NMR and IR spectra are identical with those of **7d**.

$[\alpha]_{\text{D}}^{20}$ –16.9 (*c* 1.2, MeOH).

Anal. Calcd for $\text{C}_{25}\text{H}_{33}\text{NO}_4$: C, 72.96; H, 8.08; N, 3.40. Found: C, 72.76; H, 8.28; N, 3.20.

tert-Butyl (4S)-2,2-Dimethyl-4-[3-(quinolin-3-yl)propyl]oxazolidine-3-carboxylate (7e)

Following general procedure 1 using 3-bromoquinoline (1.54 g, 995 μL) and column chromatography (*n*-hexane–EtOAc, 3:1) gave **7e** as a colorless oil; yield: 2.38 g (87%); R_f = 0.6 (*n*-hexane–EtOAc, 1:1).

$[\alpha]_{\text{D}}^{20}$ +16.2 (*c* 0.6, MeOH).

IR (film): 2978 (s), 2935 (s), 2865 (m), 1686 (s), 1570 (w), 1496 (m), 1458 (m), 1388 (s), 1257 (s), 1206 (m), 1175 (m), 1101 (s), 1075 (s), 984 (w), 908 (w), 847 (m), 788 (m), 753 (m), 614 cm^{-1} (m).

^1H NMR (250 MHz, $\text{DMSO}-d_6$): δ = 8.80 (d, J = 2.3 Hz, 1 H, H_{aryl}), 8.14 (d, J = 1.8 Hz, 1 H, H_{aryl}), 7.98 (d, J = 8.5 Hz, 1 H, H_{aryl}), 7.90 (d, J = 8 Hz, 1 H, H_{aryl}), 7.69 (m, 1 H, H_{aryl}), 7.57 (m, 1 H, H_{aryl}), 3.91–3.72 (m, 3 H, NCHCH₂O), 2.81 (m, 2 H, ArCH_2), 1.75–1.52 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$), 1.45–1.37 (m, 10 H, CH_3), 1.20 (s, 5 H, CH_3).¹⁵

^{13}C NMR (62.9 MHz, $\text{DMSO}-d_6$): δ = 151.8, 150.9, 146.2, 134.8, 133.7, 128.5, 128.4, 127.6, 127.4, 126.4, 92.5, 78.9, 78.3, 66.3, 66.0, 56.5, 32.0, 27.8, 27.0, 26.5, 24.3, 22.9.¹⁵

UV (EtOH–H₂O, 2:1): λ_{max} (ϵ) = 229 nm (39900).

Anal. Calcd for $\text{C}_{22}\text{H}_{30}\text{N}_2\text{O}_3$: C, 71.32; H, 8.16; N, 7.56. Found: C, 71.06; H, 8.13; N, 7.29.

tert-Butyl (4S)-2,2-Dimethyl-4-[3-(pyrimidin-2-yl)propyl]oxazolidine-3-carboxylate (7f)

Following general procedure 1 using 2-bromopyrimidine (1.17 g) and column chromatography (*n*-hexane–EtOAc, 3:1) gave **7f** as a colorless oil; yield: 2.05 g (86%); R_f = 0.3 (*n*-hexane–EtOAc, 1:1).

$[\alpha]_{\text{D}}^{20}$ +31.5 (*c* 1.2, MeOH).

IR (film): 2978 (s), 2936 (s), 2870 (m), 1695 (s), 1562 (s), 1478 (m), 1425 (s), 1390 (s), 1314 (m), 1257 (s), 1207 (m), 1175 (s), 1092 (s), 944 (w), 849 (m), 806 cm^{-1} (m).

^1H NMR (250 MHz, $\text{DMSO}-d_6$): δ = 8.71 (d, J = 4.8 Hz, 2 H, $\text{Hpym}_{4/6}$), 7.32 (t, J = 4.8 Hz, 1 H, Hpym_5), 3.91–3.68 (m, 3 H, NCHCH₂O), 2.87 (m, 2 H, ArCH_2), 1.69 (m, 2 H, $\text{Ar}(\text{CH}_2)_2\text{CH}_2$), 1.45–1.33 [m, 17 H, ArCH_2CH_2 , $\text{NC}(\text{CH}_3)_2$, $\text{C}(\text{CH}_3)_3$].¹⁵

^{13}C NMR (62.9 MHz, $\text{DMSO}-d_6$): δ = 170.0, 157.0, 151.0, 118.8, 92.5, 78.9, 78.4, 66.4, 56.5, 32.7, 32.1, 27.9, 26.5, 24.3, 22.9.¹⁵

UV (EtOH–H₂O, 2:1): λ_{max} (ϵ) = 248 nm (3400).

Anal. Calcd for $\text{C}_{17}\text{H}_{27}\text{N}_3\text{O}_3$: C, 63.53; H, 8.47; N, 13.07. Found: C, 63.28; H, 8.35; N, 13.12.

tert-Butyl (4R)-2,2-Dimethyl-4-[3-(pyrimidin-2-yl)propyl]oxazolidine-3-carboxylate (ent-7f)

Compound *ent-7f* was synthesized by the same procedure. NMR and IR spectra are identical with those of **7f**.

$[\alpha]_{\text{D}}^{20}$ –31.5 (*c* 1.2, MeOH).

Anal. Calcd for $\text{C}_{17}\text{H}_{27}\text{N}_3\text{O}_3$: C, 63.53; H, 8.47; N, 13.07. Found: C, 63.57; H, 8.61; N, 13.19.

tert-Butyl (4S)-4-[3-(Anthracen-9-yl)propyl]-2,2-dimethyloxazolidine-3-carboxylate (7g)

Following general procedure 1 using 9-bromoanthracene (1.90 g) and column chromatography (*n*-hexane–EtOAc, 25:1 then 10:1) gave **7g** as a highly viscous oil; yield: 2.48 g (80%); R_f = 0.6 (*n*-hexane–EtOAc, 5:1).

$[\alpha]_{\text{D}}^{20}$ +129.0 (*c* 0.5, MeOH).

IR (film): 2977 (m), 2934 (m), 1696 (s), 1624 (w), 1458 (w), 1389 (s), 1364 (s), 1259 (m), 1174 (m), 1101 (m), 1078 (m), 880 (w), 843 (w), 768 (w), 731 (m), 633 (w), 602 cm^{-1} (w).

^1H NMR (250 MHz, $\text{DMSO}-d_6$): δ = 8.47 (s, 1 H, H_{aryl}), 8.31 (d, J = 8.5 Hz, 2 H, H_{aryl}), 8.07 (m, 2 H, H_{aryl}), 7.57–7.47 (m, 4 H, H_{aryl}), 3.90–3.62 (m, 5 H, NCHCH₂O, ArCH_2), 1.74–1.56 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$), 1.47–1.34 (m, 10 H, CH_3), 1.16 (s, 5 H, CH_3).¹⁵

^{13}C NMR (100.6 MHz, $\text{DMSO}-d_6$): δ = 150.9, 134.5, 131.0, 129.0, 128.9, 125.5, 125.3, 124.8, 124.2, 92.5, 79.0, 78.3, 66.5, 56.6, 33.3, 32.9, 27.9, 27.6, 26.6, 26.4, 24.3, 22.9.¹⁵

Anal. Calcd for $\text{C}_{27}\text{H}_{33}\text{NO}_3$: C, 77.29; H, 7.93; N, 3.34. Found: C, 77.42; H, 8.07; N, 3.29.

2-Methoxy-6-(2-methoxyquinolin-6-yl)quinoline (11)

Following general procedure 1 using 6-iodo-2-methoxyquinoline (750 mg, 2.63 mmol), $\text{NiCl}_2(\text{PPh}_3)_2$ (172 mg, 0.26 mmol), **5** (1.07 g, 2.89 mmol), Zn powder (258 mg, 3.95 mmol), and DMF (4 mL) with column chromatography (*n*-hexane–EtOAc, 10:1) gave **11** as a colorless solid after recrystallization (MeCN); yield: 370 mg (89%); mp 218 °C; R_f = 0.55 (*n*-hexane–EtOAc, 3:1).

IR (KBr): 2988 (w), 2943 (w), 1624 (m), 1599 (s), 1471 (s), 1390 (s), 1326 (s), 1287 (m), 1257 (s), 1196 (w), 1117 (w), 1016 (m), 968 (w), 908 (w), 874 (m), 826 (m), 712 (w), 671 (w), 634 (w), 558 cm^{-1} (w).

^1H NMR (300 MHz, $\text{DMSO}-d_6$): δ = 8.33 (d + d, J = 2.1 Hz, J = 8.7 Hz, 2 H + 2 H, $\text{H}5'_{\text{aryl}}$, $\text{H}5'_{\text{aryl}}$, $\text{H}4'_{\text{aryl}}$, $\text{H}4'_{\text{aryl}}$), 8.14 (dd, J = 8.7 Hz, J = 2.1 Hz, 2 H, $\text{H}7'_{\text{aryl}}$, $\text{H}7'_{\text{aryl}}$), 7.90 (d, J = 8.7 Hz, 2 H, $\text{H}8'_{\text{aryl}}$, $\text{H}8'_{\text{aryl}}$), 7.08 (d, J = 8.7 Hz, 2 H, $\text{H}3'_{\text{aryl}}$, $\text{H}3'_{\text{aryl}}$), 4.02 (s, 6 H, CH_3).

^{13}C NMR (75.4 MHz, $\text{DMSO}-d_6$): δ = 162.0, 145.2, 139.4, 135.0, 128.5, 127.2, 125.3, 125.0, 113.3, 53.0.

MS (ESI): m/z (%) = 317.2 (100) [$\text{M} + \text{H}$]⁺.

Anal. Calcd for $\text{C}_{20}\text{H}_{16}\text{N}_2\text{O}_2$: C, 75.93; H, 5.10; N, 8.86. Found: C, 75.78; H, 5.15; N, 8.73.

9H-Fluoren-9-ylmethyl N-(5-Aryl-1-hydroxypentan-2-yl)carbamates 8a–g; General Procedure 2

In a round bottom flask the Negishi coupling product **7** (1.0 equiv) was treated with TFA (neat) and stirred at r.t. for 1 h. The mixture was cooled to 0 °C and aq 2 M NaOH was added until the pH of the soln became basic. The mixture was extracted with EtOAc (3 $\times$). The combined organic phases were dried (MgSO_4) and concentrated in vacuo. The residue was dissolved in EtOAc–EtOH (2:1) and Fmoc-OSu (1.05 equiv) was added and the mixture was stirred at r.t. for at least 2 h. The precipitate was filtered and washed with ice-

cold EtOAc–EtOH. Recrystallization (MeCN) gave a colorless solid.

9H-Fluoren-9-ylmethyl N-[(2S)-1-Hydroxy-5-(pyren-1-yl)pentan-2-yl]carbamate (8a)

Following general procedure 2 using **7a** (950 mg, 2.14 mmol), TFA (4 mL), Fmoc-OSu (759 mg, 2.25 mmol), and EtOAc–EtOH (2:1, 10 mL); yield: 1.03 g (91%); mp 187 °C; R_f = 0.3 (*n*-hexane–EtOAc, 1:1).

$[\alpha]_D^{20}$ +30.5 (*c* 1.0, DMF).

IR (KBr): 3302 (m), 3039 (w), 2933 (m), 2857 (w), 1686 (s), 1547 (s), 1465 (m), 1449 (m), 1352 (w), 1299 (m), 1276 (m), 1244 (m), 1181 (w), 1134 (w), 1105 (m), 1085 (m), 1051 (m), 839 (s), 819 (w), 738 (s), 680 (w), 621 cm^{−1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 8.34 (d, *J* = 9.3 Hz, 1 H, H_{aryl}), 8.27–8.20 (m, 3 H, H_{aryl}), 8.16–8.02 (m, 4 H, H_{aryl}), 7.94 (d, *J* = 7.8 Hz, 1 H, H_{aryl}), 7.84 (d, *J* = 7.5 Hz, 2 H, H_{aryl}), 7.66 (d, *J* = 7.3 Hz, 2 H, H_{aryl}), 7.38–7.16 (m, 4 H, H_{aryl}), 7.08 (d, *J* = 8.8 Hz, 1 H, NH, D₂O exch.), 4.65 (t, *J* = 5.5 Hz, 1 H, OH, D₂O exch.), 4.35–4.15 (m, 3 H, CO₂CH₂CH), 3.55 (m, 1 H, NCH), 3.48–3.22 (m, 4 H, CH₂OH, ArCH₂), 1.75 (m, 3 H, ArCH₂CH₂CH_a), 1.48 (m, 1 H, ArCH₂CH₂CH_b).

¹³C NMR (75.4 MHz, DMSO-*d*₆): δ = 155.4, 143.8, 140.6, 137.0, 130.8, 130.3, 129.1, 127.9, 127.3, 127.0, 126.8, 126.3, 125.9, 125.0, 124.8, 124.7, 124.6, 124.1, 124.0, 123.4, 119.9, 77.2, 63.7, 51.8, 46.7, 32.5, 31.0, 28.1.

Anal. Calcd for C₃₆H₃₁NO₃: C, 82.26; H, 5.94; N, 2.66. Found: C, 82.31; H, 5.98; N, 2.92.

9H-Fluoren-9-ylmethyl N-[(2R)-1-Hydroxy-5-(pyren-1-yl)pentan-2-yl]carbamate (ent-8a)

Compound *ent*-**8a** was synthesized by the same procedure. NMR and IR spectra are identical with those of **8a**.

$[\alpha]_D^{20}$ −30.5 (*c* 1.0, DMF).

9H-Fluoren-9-ylmethyl N-[(2S)-1-Hydroxy-5-(6-methoxynaphthalen-2-yl)pentan-2-yl]carbamate (8b)

Following general procedure 2 using **7b** (500 mg, 1.25 mmol), TFA (1 mL), Fmoc-OSu (443 mg, 1.31 mmol), and EtOAc–EtOH (2:1, 10 mL); yield: 470 mg (78%); mp 149 °C; R_f = 0.4 (*n*-hexane–EtOAc, 1:1).

$[\alpha]_D^{20}$ +3.8 (*c* 0.5, DMF).

IR (KBr): 3320 (m), 3060 (w), 2939 (m), 1691 (s), 1634 (w), 1607 (m), 1540 (m), 1483 (m), 1450 (m), 1391 (w), 1264 (m), 1228 (m), 1160 (w), 1087 (w), 1033 (m), 852 (w), 810 (w), 759 (m), 740 (m), 670 (w), 621 cm^{−1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 7.88 (d, *J* = 7.3 Hz, 2 H, H_{aryl}), 7.73–7.68 (m, 4 H, H_{aryl}), 7.58 (s, 1 H, H_{aryl}), 7.42–7.35 (m, 2 H, H_{aryl}), 7.32–7.23 (m, 4 H, H_{aryl}), 7.11 (dd, *J* = 8.8 Hz, *J* = 2.5 Hz, 1 H, H_{aryl}), 7.04 (d, *J* = 8.8 Hz, 1 H, NH, D₂O exch.), 4.61 (t, *J* = 5.5 Hz, 1 H, OH, D₂O exch.), 4.37–4.17 (m, 3 H, CO₂CH₂CH), 3.85 (s, 3 H, OCH₃), 3.48 (m, 1 H, NCH), 3.29 (m, 2 H, CH₂OH), 2.69 (m, 2 H, ArCH₂), 1.62 (m, 3 H, ArCH₂CH₂CH_a), 1.34 (m, 1 H, ArCH₂CH₂CH_b).

¹³C NMR (75.4 MHz, DMSO-*d*₆): δ = 156.6, 155.8, 143.8, 140.6, 137.2, 132.5, 128.6, 128.5, 127.6, 127.4, 126.8, 126.5, 125.7, 125.1, 119.9, 118.2, 105.7, 65.0, 63.5, 55.0, 52.6, 46.7, 34.9, 30.5, 27.4.

Anal. Calcd for C₃₁H₃₁NO₄: C, 77.31; H, 6.49; N, 2.91. Found: C, 77.16; H, 6.32; N, 2.93.

9H-Fluoren-9-ylmethyl N-[(2S)-1-Hydroxy-5-(phenanthren-9-yl)pentan-2-yl]carbamate (8c)

Following general procedure 2 using **7c** (1.13 g, 2.69 mmol), TFA

(4 mL), Fmoc-OSu (954 mg, 2.83 mmol), and EtOAc–EtOH (2:1, 15 mL); yield: 1.19 g (88%); mp 182 °C; R_f = 0.25 (*n*-hexane–EtOAc, 1:1).

$[\alpha]_D^{20}$ −2.4 (*c* 0.5, DMF).

IR (KBr): 3321 (m), 3063 (w), 2947 (w), 2865 (w), 1690 (s), 1543 (m), 1449 (m), 1276 (m), 1243 (m), 1147 (w), 1088 (w), 1043 (m), 999 (w), 888 (w), 759 (m), 740 (s), 671 (w), 619 cm^{−1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 8.86 (d, *J* = 7.8 Hz, 1 H, H_{aryl}), 8.78 (d, *J* = 8.3 Hz, 1 H, H_{aryl}), 8.14 (d, *J* = 7.3 Hz, 1 H, H_{aryl}), 7.91–7.84 (m, 3 H, H_{aryl}), 7.68–7.60 (m, 7 H, H_{aryl}), 7.39–7.18 (m, 4 H, H_{aryl}), 7.08 (d, *J* = 8.5 Hz, 1 H, NH, D₂O exch.), 4.64 (t, *J* = 5.3 Hz, 1 H, OH, D₂O exch.), 4.35–4.16 (m, 3 H, CO₂CH₂CH), 3.55 (m, 1 H, NCH), 3.35 (m, 2 H, CH₂OH), 3.09 (m, 2 H, ArCH₂), 1.75 (m, 3 H, ArCH₂CH₂CH_a), 1.49 (m, 1 H, ArCH₂CH₂CH_b).

¹³C NMR (75.4 MHz, DMSO-*d*₆): δ = 155.9, 143.8, 143.7, 140.5, 136.4, 131.3, 130.5, 130.0, 128.9, 127.7, 127.4, 126.8, 126.6, 126.2, 126.0, 125.6, 125.0, 124.2, 123.2, 122.5, 119.9, 65.0, 63.6, 52.5, 46.7, 32.5, 30.9, 26.4.

Anal. Calcd for C₃₄H₃₁NO₃: C, 81.41; H, 6.23; N, 2.79. Found: C, 81.44; H, 6.23; N, 2.84.

9H-Fluoren-9-ylmethyl N-[(2S)-1-Hydroxy-5-(4-phenoxyphenyl)pentan-2-yl]carbamate (8d)

Following general procedure 2 using **7d** (1.75 g, 4.25 mmol), TFA (5 mL), Fmoc-OSu (1.51 g, 4.46 mmol), and EtOAc–EtOH (2:1, 15 mL); yield: 1.57 g (75%); mp 94–96 °C; R_f = 0.4 (EtOAc–*n*-hexane, 1:1).

$[\alpha]_D^{20}$ −2.0 (*c* 0.5, DMF).

IR (KBr): 3335 (m), 3062 (w), 2938 (m), 2858 (m), 1688 (s), 1590 (m), 1535 (s), 1506 (m), 1488 (m), 1449 (m), 1321 (w), 1241 (s), 1164 (w), 1087 (m), 1042 (m), 978 (w), 870 (w), 830 (w), 737 (m), 691 cm^{−1} (m).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 7.88 (d, *J* = 7.5 Hz, 2 H, H_{aryl}), 7.69 (d, *J* = 7.5 Hz, 2 H, H_{aryl}), 7.42–7.20 (m, 6 H, H_{aryl}), 7.16–6.89 (m, 8 H, H_{aryl}, NH, D₂O exch.), 4.63 (t, *J* = 5.5 Hz, 1 H, OH, D₂O exch.), 4.36–4.17 (m, 3 H, CO₂CH₂CH), 3.46 (m, 1 H, NCH), 3.38–3.20 (m, 2 H, CH₂), 2.52 (m, 2 H, CH₂), 1.54 (m, 3 H, ArCH₂CH₂CH_a), 1.30 (m, 1 H, ArCH₂CH₂CH_b).

¹³C NMR (75.4 MHz, DMSO-*d*₆): δ = 156.9, 155.8, 154.3, 143.8, 140.6, 137.3, 129.8, 129.5, 127.4, 126.8, 125.0, 122.9, 119.9, 118.5, 118.0, 65.0, 63.4, 52.5, 46.7, 34.2, 30.4, 27.5.

Anal. Calcd for C₃₂H₃₁NO₄: C, 77.87; H, 6.33; N, 2.84; found: C, 77.61; H, 6.45; N, 2.92.

9H-Fluoren-9-ylmethyl N-[(2S)-1-Hydroxy-5-(quinolin-3-yl)pentan-2-yl]carbamate (8e)

Following general procedure 2 using **7e** (1.13 g, 3.05 mmol), TFA (5 mL), Fmoc-OSu (1.08 g, 3.20 mmol), and EtOAc–EtOH (2:1, 15 mL); yield: 1.17 g (85%); mp 127–128 °C; R_f = 0.2 (EtOAc–*n*-hexane, 10:1).

$[\alpha]_D^{20}$ −2.0 (*c* 0.5, DMF).

IR (KBr): 3420 (s), 3319 (s), 3062 (w), 2929 (m), 1690 (s), 1542 (m), 1450 (m), 1334 (w), 1244 (m), 1051 (m), 740 cm^{−1} (m).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 8.78 (d, *J* = 2 Hz, 1 H, H_{aryl}), 8.10 (br s, 1 H, H_{aryl}), 7.99 (d, *J* = 8.3 Hz, 1 H, H_{aryl}), 7.88 (m, 3 H, H_{aryl}), 7.70 (m, 3 H, H_{aryl}), 7.57 (m, 1 H, H_{aryl}), 7.37 (m, 2 H, H_{aryl}), 7.27 (m, 2 H, H_{aryl}), 7.04 (d, *J* = 8.8 Hz, 1 H, NH, D₂O exch.), 4.62 (t, *J* = 5.5 Hz, 1 H, OH, D₂O exch.), 4.37–4.17 (m, 3 H, CO₂CH₂CH), 3.49 (m, 1 H, NCH), 3.30 (m, 2 H, CH₂OH), 2.78 (m, 2 H, ArCH₂), 1.64 (m, 3 H, ArCH₂CH₂CH_a), 1.35 (m, 1 H, ArCH₂CH₂CH_b).

^{13}C NMR (75.4 MHz, DMSO- d_6): δ = 157.6, 151.8, 146.1, 143.8, 142.5, 140.6, 139.3, 137.3, 135.0, 133.6, 128.8, 128.5, 128.4, 127.6, 127.4, 127.1, 126.8, 126.3, 125.0, 121.2, 119.9, 109.5, 63.9, 52.1, 46.7, 32.1, 30.5, 27.1.

Anal. Calcd for $\text{C}_{29}\text{H}_{28}\text{N}_2\text{O}_3$: C, 76.97; H, 6.24; N, 6.19. Found: C, 77.23; H, 6.30; N, 6.22.

9H-Fluoren-9-ylmethyl N-[(2S)-1-Hydroxy-5-(pyrimidin-2-yl)pentan-2-yl]carbamate (8f)

In a round-bottom flask **7f** (1.80 g, 5.60 mmol) was treated with TFA (5 mL) and stirred at r.t. for 1 h. The mixture was concentrated and dried in vacuo, then the residue was dissolved in EtOAc–EtOH (2:1, 15 mL). Fmoc-OSu (2.08 g, 6.16 mmol) and DIPEA (2.88 mL, 16.8 mmol) were added and the mixture was stirred at r.t. for at least 2 h. Brine (30 mL) was added and the mixture was extracted with EtOAc (3 $\times$ 50 mL). The combined organic phases were dried (MgSO_4), concentrated under reduced pressure, and adsorbed on silica gel. Purification by column chromatography (EtOAc–*n*-hexane, 1:1, EtOAc, then EtOAc–MeOH, 9:1) and recrystallization (MeCN) gave **8f** (2.12 g, 94%) as a colorless solid; mp 73–75 °C; R_f = 0.35 (EtOAc–MeOH, 9:1).

$[\alpha]_{\text{D}}^{20}$ –6.7 (*c* 0.6, MeOH).

IR (KBr): 3317 (m), 3039 (w), 2947 (m), 1689 (s), 1561 (s), 1541 (s), 1450 (m), 1423 (m), 1245 (m), 1139 (w), 1085 (m), 1054 (m), 759 (m), 737 (m), 636 cm^{-1} (w).

^1H NMR (250 MHz, DMSO- d_6): δ = 8.70 (d, J = 4.8 Hz, 2 H, H_{aryl}), 7.88 (d, J = 7.5 Hz, 2 H, H_{aryl}), 7.70 (d, J = 7.3 Hz, 2 H, H_{aryl}), 7.40 (t, J = 7.3 Hz, 2 H, H_{aryl}), 7.34–7.26 (m, 3 H, H_{aryl}), 7.04 (d, J = 8.5 Hz, 1 H, NH, D_2O exch.), 4.61 (t, J = 5.5 Hz, 1 H, OH, D_2O exch.), 4.35–4.17 (m, 3 H, $\text{CO}_2\text{CH}_2\text{CH}$), 3.46 (m, 1 H, NCH), 3.29 (m, 2 H, CH_2OH), 2.85 (m, 2 H, ArCH_2), 1.67 (m, 3 H, $\text{ArCH}_2\text{CH}_2\text{CH}_a$), 1.34 (m, 1 H, $\text{ArCH}_2\text{CH}_2\text{CH}_b$).

^{13}C NMR (100.6 MHz, DMSO- d_6): δ = 170.2, 157.0, 155.9, 143.9, 140.6, 127.5, 126.9, 125.2, 120.0, 118.8, 65.1, 63.5, 52.7, 46.7, 38.6, 30.4, 24.6.

Anal. Calcd for $\text{C}_{24}\text{H}_{25}\text{N}_3\text{O}_3$: C, 71.44; H, 6.25; N, 10.41. Found: C, 71.23; H, 6.30; N, 10.34.

9H-Fluoren-9-ylmethyl N-[(2R)-1-Hydroxy-5-(pyrimidin-2-yl)pentan-2-yl]carbamate (ent-8f)

Compound *ent*-**8f** was synthesized by the same procedure. NMR and IR spectra are identical with those of **8f**.

$[\alpha]_{\text{D}}^{20}$ +6.7 (*c* 0.6, MeOH).

MS (ESI): m/z (%) = 404.1 (100) [$\text{M} + \text{H}$] $^+$.

9H-Fluoren-9-ylmethyl N-[(2S)-5-(Anthracen-9-yl)-1-hydroxy-pentan-2-yl]carbamate (8g)

Following general procedure 2 using **7g** (500 mg, 1.19 mmol), TFA (2 mL), Fmoc-OSu (422 mg, 1.25 mmol), and EtOAc–EtOH (2:1, 10 mL) gave **8g** as a colorless solid; yield: 510 mg (85%); mp 185–186 °C; R_f = 0.4 (EtOAc–*n*-hexane, 1:1).

$[\alpha]_{\text{D}}^{20}$ –1.6 (*c* 0.3, DMF).

IR (KBr): 3424 (s), 3058 (m), 2947 (m), 1687 (s), 1544 (m), 1448 (m), 1384 (m), 1352 (m), 1301 (m), 1277 (m), 1240 (m), 1104 (w), 1085 (m), 1056 (m), 1021 (w), 878 (w), 836 (w), 789 (w), 758 (m), 738 (s), 560 cm^{-1} (w).

^1H NMR (400 MHz, DMSO- d_6): δ = 8.47 (s, 1 H, H_{aryl}), 8.31 (m, 2 H, H_{aryl}), 8.07 (m, 2 H, H_{aryl}), 7.82 (t, J = 7.2 Hz, 2 H, H_{aryl}), 7.63 (d, J = 7.2 Hz, 2 H, H_{aryl}), 7.50 (m, 4 H, H_{aryl}), 7.32 (t, J = 7.2 Hz, 2 H, H_{aryl}), 7.20 (t, J = 7.2 Hz, 1 H, H_{aryl}), 7.11 (m, 2 H, H_{aryl}), NH, D_2O exch.), 4.69 (t, J = 5.6 Hz, 1 H, OH, D_2O exch.), 4.27 (m, 2 H, $\text{CO}_2\text{CH}_2\text{CH}$), 4.16 (m, 1 H, $\text{CO}_2\text{CH}_2\text{CH}$), 3.67–3.49 (m, 3 H, NCH,

CH_2OH), 3.33 (m, 2 H, ArCH_2), 1.86 (m, 1 H, ArCH_2CH_a), 1.71 (m, 2 H, $\text{ArCH}_2\text{CH}_2\text{CH}_a$), 1.58 (m, 1 H, $\text{ArCH}_2\text{CH}_2\text{CH}_b$).

^{13}C NMR (75.4 MHz, DMSO- d_6): δ = 155.9, 143.8, 143.7, 140.5, 134.8, 131.0, 128.9, 128.8, 127.4, 126.8, 125.5, 125.1, 124.9, 124.2, 119.9, 65.0, 63.8, 52.5, 46.7, 31.1, 27.6, 27.1.

Anal. Calcd for $\text{C}_{34}\text{H}_{31}\text{NO}_3$: C, 81.41; H, 6.23; N, 2.79. Found: C, 81.13; H, 6.32; N, 2.87.

5-Aryl-2-[(9H-fluoren-9-ylmethoxy)carbonyl]amino}pentanoic Acid; General Procedure 3

The Fmoc-amino alcohol **8** (1.0 equiv) was dissolved in DMF and PDC was added. [In the case of **8e** and **8d**, H_5IO_6 was added.] The mixture was stirred 40 °C for 8 h and then it was quenched with brine (40 mL) and 6 M HCl (few drops). The soln was extracted with EtOAc (3 $\times$ 50 mL). The combined organic phases were washed with sat. $\text{Na}_2\text{S}_2\text{O}_5$ soln, dried (MgSO_4), and concentrated in vacuo. The residue was purified by column chromatography (EtOAc–*n*-hexane, 1:1, EtOAc, then EtOAc–MeOH, 9:1). Recrystallization (CH_2Cl_2 –*n*-hexane) gave a solid.

(2S)-2-[(9H-Fluoren-9-ylmethoxy)carbonyl]amino}-5-(pyren-1-yl)pentanoic Acid (9a)

Following general procedure 3 using **8a** (350 mg, 0.67 mmol), PDC (1.25 g, 3.33 mmol), and DMF (5 mL) gave **9a** as a light brown solid; yield: 299 mg (83%); mp 115–116 °C; >99% ee [HPLC analysis (Chiral DAICEL OJ-RH column, 0.1% TFA–MeCN, 42:58, 0.5 mL/min, 254 nm): t_R = 35.28 (*R*), 38.51 min (*S*)]; R_f = 0.25 (EtOAc–MeOH, 9:1).

$[\alpha]_{\text{D}}^{20}$ +1.0 (*c* 1.0, DMF).

IR (KBr): 3424 (s), 3039 (w), 2936 (w), 1703 (s), 1655 (m), 1636 (m), 1562 (m), 1525 (m), 1458 (m), 1421 (m), 1349 (m), 1246 (m), 1055 (m), 844 (m), 740 cm^{-1} (m).

^1H NMR (400 MHz, DMSO- d_6): δ = 12.59 (br s, 1 H, CO_2H , D_2O exch.), 8.42–8.13 (m, 7 H, H_{aryl}), 8.03 (m, 1 H, H_{aryl}), 7.94 (d, J = 7.6 Hz, 1 H, H_{aryl}), 7.86 (d, J = 7.2 Hz, 2 H, H_{aryl}), 7.70 (m, 3 H, H_{aryl}), NH, D_2O exch.), 7.35 (m, 2 H, H_{aryl}), 7.24 (m, 2 H, H_{aryl}), 4.29–4.18 (m, 3 H, $\text{CO}_2\text{CH}_2\text{CH}$), 4.07 (m, 1 H, NCH), 3.35 (m, 2 H, ArCH_2), 1.85 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$).

^{13}C NMR (75.4 MHz, DMSO- d_6): δ = 173.8, 156.0, 143.7, 143.6, 140.6, 136.5, 130.8, 130.3, 129.1, 128.0, 127.4, 127.3, 127.0, 126.9, 126.3, 126.0, 125.1, 124.8, 124.6, 124.1, 124.0, 123.4, 119.9, 65.4, 53.7, 46.5, 32.1, 30.9, 28.0.

MS (MALDI): m/z (%) = 540.05 (100) [$\text{M} + \text{H}$] $^+$, 495.87 (93) [$\text{M} + \text{H} - \text{CO}_2$] $^+$, 317.29 (15) [$\text{M} + \text{H} - \text{Fmoc}$] $^+$.

(2R)-2-[(9H-Fluoren-9-ylmethoxy)carbonyl]amino}-5-(pyren-1-yl)pentanoic Acid (ent-9a)

Compound *ent*-**9a** was synthesized by the same procedure. NMR and IR spectra are identical with those of **9a**.

$[\alpha]_{\text{D}}^{20}$ –1.0 (*c* 1.0, DMF).

(2S)-2-[(9H-Fluoren-9-ylmethoxy)carbonyl]amino}-5-(6-methoxynaphthalen-2-yl)pentanoic Acid (9b)

Following general procedure 3 using **8b** (350 mg, 0.73 mmol), PDC (1.64 g, 5.09 mmol), and DMF (5 mL) gave **9b** as a light brown solid; yield: 287 mg (80%); mp 101 °C; R_f = 0.3 (EtOAc–MeOH, 9:1).

$[\alpha]_{\text{D}}^{20}$ +2.0 (*c* 0.45, DMF).

IR (KBr): 3421 (s), 3063 (m), 2938 (m), 1688 (s), 1599 (s), 1525 (m), 1479 (m), 1449 (s), 1419 (s), 1350 (m), 1267 (s), 1159 (m), 1063 (m), 855 (w), 808 (w), 760 (m), 741 (m), 621 cm^{-1} (w).

^1H NMR (250 MHz, DMSO- d_6): δ = 12.52 (br s, 1 H, CO_2H , D_2O exch.), 7.88 (d, J = 7.5 Hz, 2 H, H_{aryl}), 7.75–7.65 (m, 5 H, H_{aryl}), NH, D_2O exch.), 7.59 (s, 1 H, H_{aryl}), 7.38 (m, 2 H, H_{aryl}), 7.30–7.26 (m,

^1H NMR (250 MHz, DMSO- d_6): δ = 7.12 (dd, J = 9 Hz, J = 2.3 Hz, 1 H, H_{aryl}), 4.30–4.18 (m, 3 H, $\text{CO}_2\text{CH}_2\text{CH}$), 4.00 (m, 1 H, NCH), 3.85 (s, 3 H, OCH_3), 2.70 (m, 2 H, ArCH_2), 1.71 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$).

^{13}C NMR (75.4 MHz, DMSO- d_6): δ = 173.7, 156.6, 156.0, 143.7, 140.6, 136.8, 132.6, 128.6, 128.4, 127.6, 127.4, 126.9, 126.5, 125.8, 125.1, 119.9, 118.3, 105.7, 65.4, 55.0, 53.5, 46.5, 34.4, 30.4, 27.2.

MS (ESI): m/z (%) = 272.0 (100) $[\text{M} - \text{H} - \text{Fmoc}]^-$, 298.0 (33.62) $[\text{M} - \text{H} - \text{fluorenylmethanol}]^-$, 494.3 (72.69) $[\text{M} - \text{H}]^-$, 530.3 (22.73) $[\text{M} + ^{35}\text{Cl}]^-$, 532.3 (11.52) $[\text{M} + ^{37}\text{Cl}]^-$.

(2S)-2-[[[(9H-Fluoren-9-ylmethoxy)carbonyl]amino]-5-(phenanthren-9-yl)pentanoic Acid (9c)

Following general procedure 3 using **8c** (180 mg, 0.36 mmol), PDC (1.35 g, 3.60 mmol), and DMF (4 mL) gave **9c** as a light brown solid; yield: 171 mg (92%); mp 119–120 °C; R_f = 0.25 (EtOAc–MeOH, 9:1).

$[\alpha]_{\text{D}}^{20}$ –3.0 (c 0.4, DMF).

IR (KBr): 3422 (s), 3066 (m), 2950 (m), 2867 (m), 1700 (s), 1638 (m), 1542 (m), 1527 (m), 1449 (m), 1423 (m), 1343 (m), 1244 (m), 1081 (m), 1053 (m), 947 (w), 892 (w), 741 cm^{-1} (s).

^1H NMR (250 MHz, DMSO- d_6): δ = 8.85 (m, 1 H, H_{aryl}), 8.78 (m, 1 H, H_{aryl}), 8.13 (d, J = 8.0 Hz, 1 H, H_{aryl}), 7.86 (m, 3 H, H_{aryl}), 7.68–7.56 (m, 7 H, H_{aryl}), 7.36 (m, 2 H, H_{aryl}), 7.25 (m, 2 H, H_{aryl}), 7.03 (br s, 1 H, NH), 4.29–4.16 (m, 3 H, $\text{CO}_2\text{CH}_2\text{CH}$), 3.87 (m, 1 H, NCH), 3.07 (m, 2 H, ArCH_2), 1.92–1.63 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$). The CO_2H signal was not observed.

^{13}C NMR (75.4 MHz, DMSO- d_6): δ = 155.5, 143.8, 140.5, 136.4, 131.3, 130.6, 130.0, 128.9, 127.7, 127.4, 126.9, 126.6, 126.2, 126.0, 125.5, 125.1, 125.0, 124.3, 123.3, 122.5, 119.9, 65.2, 54.8, 46.6, 32.4, 32.1, 26.2. The CO_2H signal was not observed.

MS (MALDI): m/z (%) = 554.91 (100) $[\text{M} + \text{K}]^+$.

(2S)-2-[[[(9H-Fluoren-9-ylmethoxy)carbonyl]amino]-5-(4-phenoxyphenyl)pentanoic Acid (9d)

Following general procedure 3 using **8d** (200 mg, 0.41 mmol), PDC (150 mg, 0.41 mmol), H_5IO_6 (547 mg, 2.43 mmol), and DMF (3 mL) gave **9d** as a colorless solid; yield: 144 mg (70%); mp 59–60 °C; R_f = 0.3 (EtOAc–MeOH, 9:1).

$[\alpha]_{\text{D}}^{20}$ –1.5 (c 0.4, DMF).

IR (KBr): 3402 (m), 3036 (m), 2924 (s), 2854 (m), 1719 (s), 1590 (m), 1507 (s), 1489 (s), 1450 (m), 1417 (w), 1340 (w), 1238 (s), 1166 (w), 1106 (w), 1057 (w), 870 (w), 741 (m), 692 (w), 620 cm^{-1} (w).

^1H NMR (250 MHz, DMSO- d_6): δ = 12.55 (br s, 1 H, CO_2H , D_2O exch.), 7.89 (d, J = 7.3 Hz, 2 H, H_{aryl}), 7.74–7.65 (m, 3 H, H_{aryl} , NH, D_2O exch.), 7.44–7.27 (m, 6 H, H_{aryl}), 7.20 (d, J = 8.5 Hz, 2 H, H_{aryl}), 7.11 (t, J = 7.3 Hz, 1 H, H_{aryl}), 6.95 (m, 4 H, H_{aryl}), 4.30–4.19 (m, 3 H, $\text{CO}_2\text{CH}_2\text{CH}$), 3.97 (m, 1 H, NCH), 2.56 (m, 2 H, ArCH_2), 1.76–1.56 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$).

^{13}C NMR (75.4 MHz, DMSO- d_6): δ = 173.7, 156.9, 156.0, 154.4, 143.7, 140.6, 136.9, 129.8, 129.6, 127.5, 126.9, 125.1, 123.0, 120.0, 118.6, 118.0, 65.4, 53.5, 46.5, 33.7, 30.3, 27.4.

Anal. Calcd for $\text{C}_{32}\text{H}_{29}\text{NO}_5$: C, 75.72; H, 5.76; N, 2.76. Found: C, 75.59; H, 5.93; N, 2.74.

(2S)-2-[[[(9H-Fluoren-9-ylmethoxy)carbonyl]amino]-5-(quinolin-3-yl)pentanoic Acid (9e)

Following general procedure 3 using **8e** (500 mg, 1.10 mmol), PDC (420 mg, 1.10 mmol), H_5IO_6 (1.00 g, 4.40 mmol), and DMF (10 mL) gave **9e** as a colorless solid; yield: 424 mg (82%); mp 103–105 °C; R_f = 0.3 (EtOAc–MeOH, 2:1).

$[\alpha]_{\text{D}}^{20}$ –3.7 (c 0.4, DMF).

IR (KBr): 3422 (m), 3328 (m), 3063 (w), 2938 (m), 2862 (w), 1700 (s), 1534 (m), 1499 (m), 1450 (m), 1388 (w), 1233 (s), 1152 (m), 1134 (m), 1084 (m), 1054 (m), 912 (w), 758 (s), 740 cm^{-1} (s).

^1H NMR (250 MHz, DMSO- d_6): δ = 12.35 (br s, 1 H, CO_2H , D_2O exch.), 8.79 (d, J = 2.3 Hz, 1 H, H_{aryl}), 8.12 (d, J = 1.3 Hz, 1 H, H_{aryl}), 7.99 (d, J = 8.3 Hz, 1 H, H_{aryl}), 7.89 (m, 3 H, H_{aryl}), 7.73–7.55 (m, 5 H, H_{aryl} , NH, D_2O exch.), 7.39 (m, 2 H, H_{aryl}), 7.29 (m, 2 H, H_{aryl}), 4.30–4.18 (m, 3 H, $\text{CO}_2\text{CH}_2\text{CH}$), 4.03 (m, 1 H, NCH), 2.80 (m, 2 H, ArCH_2), 1.76–1.60 (m, 4 H, $\text{ArCH}_2\text{CH}_2\text{CH}_2$).

^{13}C NMR (75.4 MHz, DMSO- d_6): δ = 173.7, 156.0, 151.8, 146.2, 143.7, 140.6, 134.6, 133.8, 128.5, 127.6, 127.5, 127.4, 126.9, 126.4, 125.13, 125.10, 119.9, 65.5, 53.5, 46.6, 31.6, 30.2, 26.9.

MS (ESI): m/z (%) = 687.6 (9.99) $[\text{M} - \text{H} + \text{Fmoc}]^-$, 465.1 (57.69) $[\text{M} - \text{H}]^-$, 343.1 (57.40) $[\text{M} - 2 \text{H} + \text{Fmoc}]^{2-}$, 269.0 (72.31) $[\text{M} - \text{H} - \text{fluorenylmethanol}]^-$, 243.0 (100.0) $[\text{M} - \text{H} - \text{Fmoc}]^-$.

(2S)-2-[[[(9H-Fluoren-9-ylmethoxy)carbonyl]amino]-5-(pyrimidin-2-yl)pentanoic Acid (9f)

The Fmoc-amino alcohol **8f** (620 mg, 1.54 mmol) was dissolved in acetone (10 mL). PDC (56 mg, 0.15 mmol) and H_5IO_6 (1.65 g, 7.70 mmol) were added to the clear soln. The mixture was stirred overnight at r.t. and then it was treated with brine (20 mL) and 6 M HCl (few drops). The soln was extracted with EtOAc (3 $\times$ 50 mL). The combined organic phases were washed with sat. $\text{Na}_2\text{S}_2\text{O}_5$ soln, dried (MgSO_4), and concentrated in vacuo. The residue was purified by column chromatography (EtOAc–*n*-hexane, 1:1, EtOAc, then EtOAc–MeOH, 9:1). Recrystallization (MeCN– H_2O) gave **9f** (504 mg, 79%) as a yellow solid; mp 61–63 °C; R_f = 0.2 (EtOAc–MeOH, 2:1).

$[\alpha]_{\text{D}}^{20}$ –1.9 (c 0.4, MeOH).

IR (KBr): 3339 (s), 3066 (m), 2943 (m), 1698 (s), 1543 (s), 1449 (m), 1332 (m), 1267 (s), 1199 (s), 1140 (m), 1056 (m), 834 (w), 803 (w), 761 (m), 740 (m), 648 cm^{-1} (w).

^1H NMR (400 MHz, DMSO- d_6): δ = 12.53 (br s, 1 H, CO_2H , D_2O exch.), 8.72 (d, J = 4.8 Hz, 2 H, H_{aryl}), 7.89 (d, J = 7.7 Hz, 2 H, H_{aryl}), 7.72 (d, J = 7.2 Hz, 2 H, H_{aryl}), 7.67 (d, J = 8 Hz, 1 H, NH, D_2O exch.), 7.41 (t, J = 7.2 Hz, 2 H, H_{aryl}), 7.34–7.29 (m, 3 H, H_{aryl}), 4.27 (m, 2 H, $\text{CO}_2\text{CH}_2\text{CH}$), 4.22 (m, 1 H, $\text{CO}_2\text{CH}_2\text{CH}$), 3.98 (m, 1 H, NCH), 2.87 (m, 2 H, ArCH_2), 1.83–1.73 (m, 3 H, $\text{ArCH}_2\text{CH}_2\text{CH}_a$), 1.67 (m, 1 H, $\text{ArCH}_2\text{CH}_2\text{CH}_b$).

^{13}C NMR (75.4 MHz, DMSO- d_6): δ = 173.7, 169.8, 157.0, 156.0, 143.6, 140.6, 127.5, 126.9, 125.1, 119.9, 118.8, 65.5, 53.5, 46.5, 38.0, 30.2, 24.4.

MS (ESI): m/z (%) = 418.2 (100) $[\text{M} + \text{H}]^+$.

(2R)-2-[[[(9H-Fluoren-9-ylmethoxy)carbonyl]amino]-5-(pyrimidin-2-yl)pentanoic Acid (ent-9f)

$[\alpha]_{\text{D}}^{20}$ +1.9 (c 0.4, MeOH).

(2S)-2-[[[(9H-Fluoren-9-ylmethoxy)carbonyl]amino]-5-(9-hydroxy-10-oxo-9,10-dihydroanthracen-9-yl)pentanoic Acid (12)

Following general procedure 3 using **8g** (100 mg, 0.20 mmol), PDC (750 mg, 2.00 mmol), and DMF (1.5 mL) gave **12** as a colorless solid; yield: 40 mg (37%); mp 104–106 °C; R_f = 0.5 (EtOAc–MeOH, 2:1).

$[\alpha]_{\text{D}}^{20}$ –8.9 (c 0.2, MeOH).

IR (KBr): 3407 (s), 3066 (m), 2938 (m), 1705 (s), 1656 (s), 1600 (s), 1525 (m), 1450 (m), 1420 (m), 1320 (s), 1260 (m), 1215 (m), 1174 (m), 1074 (m), 932 (m), 818 (w), 760 (m), 741 (m), 695 (m), 622 cm^{-1} (w).

^1H NMR (250 MHz, DMSO- d_6): δ = 12.40 (br s, 1 H, CO_2H , D_2O exch.), 8.06 (d, J = 7.5 Hz, 2 H, H_{aryl}), 7.91 (m, 4 H, H_{aryl}), 7.70 (m, 4 H, H_{aryl}), 7.51–7.28 (m, 7 H, H_{aryl} , NH, D_2O exch.), 6.24 (br s, 1

H, OH, D₂O exch.), 4.21–4.10 (m, 3 H, CO₂CH₂CH), 3.65 (m, 1 H, NCH), 1.88 (m, 2 H, anthrone-CH₂), 1.35 (m, 2 H, ArCH₂CH₂CH₂), 0.73 (m, 2 H, ArCH₂CH₂).

¹³C NMR (75.4 MHz, DMSO-*d*₆): δ = 182.9, 173.4, 155.8, 148.4, 143.6, 140.5, 133.4, 130.0, 127.4, 126.9, 126.5, 126.4, 125.6, 125.1, 119.9, 71.1, 65.5, 53.2, 47.9, 46.5, 30.3, 20.2.

MS (ESI): *m/z* (%) = 546.3 (100) [M – H][–], 324.0 (69.3) [M – H – Fmoc][–], 350.0 (83.9) [M – H – fluorenylmethanol][–].

Dimethyl (2S)-2-[(*tert*-Butoxycarbonyl)amino]butanedioate (14)

The synthetic procedure was analogous to that for **2**; mp 56 °C; *R*_f = 0.4 (*n*-hexane–EtOAc 1:1).

[α]_D²⁰ –17.9 (*c* 1.4, EtOH).

IR (KBr): 3408 (m), 2982 (m), 1740 (s), 1706 (s), 1508 (s), 1441 (m), 1378 (s), 1349 (s), 1296 (m), 1218 (s), 1160 (s), 1072 (w), 1032 (m), 998 (w), 869 (w), 844 (w), 782 (w), 685 (w), 592 cm^{–1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 7.30 (d, *J* = 8.3 Hz, 1 H, NH, D₂O exch.), 4.37 (m, 1 H, NCH), 3.62 (s, 3 H, CHCO₂CH₃), 3.60 (s, 3 H, CH₂CO₂CH₃), 2.66 (m, 2 H, CH₂), 1.37 [s, 9 H, C(CH₃)₃].

¹³C NMR (75.4 MHz, DMSO-*d*₆): δ = 171.4, 170.3, 155.0, 78.4, 51.9, 51.5, 49.9, 35.6, 28.0.

Anal. Calcd for C₁₁H₁₉NO₆: C, 50.57; H, 7.33; N, 5.36. Found: C, 50.58; H, 7.33; N, 5.40.

tert-Butyl *N*-[(2S)-1,4-Dihydroxybutan-2-yl]carbamate (15)

The synthetic procedure was analogous to that for **3**; mp 45 °C; *R*_f = 0.2 (EtOAc–*n*-hexane 10:1).

[α]_D²⁵ –31.0 (*c* 1.0, MeOH).

IR (KBr): 3376 (m), 2970 (m), 1693 (s), 1518 (s), 1391 (m), 1366 (s), 1296 (m), 1247 (s), 1172 (m), 1049 (s), 972 (m), 877 (m), 844 (w), 783 (w), 587 cm^{–1} (m).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 6.45 (d, *J* = 8.3 Hz, 1 H, NH, D₂O exch.), 4.55 (t, *J* = 5.5 Hz, 1 H, CHCH₂OH, D₂O exch.), 4.34 (t, *J* = 5.3 Hz, 1 H, CH₂CH₂OH, D₂O exch.), 3.46–3.19 (m, 5 H, CH, CH₂OH), 1.64 (m, 1 H, NCHCH_a), 1.45 (m, 1 H, NCHCH_b), 1.37 [s, 9 H, C(CH₃)₃].

¹³C NMR (62.9 MHz, DMSO-*d*₆): δ = 155.3, 77.3, 63.4, 57.9, 49.5, 34.3, 28.2.

Anal. Calcd for C₉H₁₉NO₄: C, 52.67; H, 9.33; N, 6.82. Found: C, 52.84; H, 9.12; N, 6.90.

tert-Butyl (4S)-4-(2-Hydroxyethyl)-2,2-dimethyloxazolidine-3-carboxylate (16)

The synthetic procedure was analogous to that for **4**; mp 44–47 °C; *R*_f = 0.6 (*n*-hexane–EtOAc 2:1).

[α]_D²⁰ +8.6 (*c* 1.0, MeOH).

IR (KBr): 3255 (m), 2979 (s), 2938 (s), 2881 (s), 1694 (s), 1482 (m), 1393 (s), 1311 (m), 1248 (m), 1220 (s), 1174 (s), 1084 (s), 1036 (m), 980 (m), 942 (w), 884 (m), 849 (w), 812 (w), 770 (m), 700 (w), 656 (w), 570 cm^{–1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 6.45 (t, *J* = 5.3 Hz, 1 H, OH, D₂O exch.), 3.87–3.54 (m, 3 H, CHCH₂O, CHCH₂O), 3.43 (m, 2 H, CH₂CH₂OH), 1.77 (m, 1 H, NCHCH_aH_b), 1.59 (m, 1 H, NCHCH_aH_b), 1.46–1.38 (m, 14 H, CH₃), 1.22 (m, 1 H, CH₃).¹⁵

¹³C NMR (75.4 MHz, DMSO-*d*₆): δ = 154.7, 151.0, 127.9, 126.5, 126.1, 100.4, 99.2, 92.2, 78.9, 78.5, 77.6, 66.7, 66.4, 63.7, 61.3, 58.2, 57.7, 55.0, 49.3, 47.6, 39.4, 36.4, 35.5, 34.9, 28.1, 27.9, 27.3, 26.5, 24.6, 24.5, 24.3, 24.0, 22.9.¹⁵

Anal. Calcd for C₁₂H₂₃NO₄: C, 58.75; H, 9.45; N, 5.71. Found: C, 58.76; H, 9.34; N, 5.70.

tert-Butyl (4S)-4-(2-Iodoethyl)-2,2-dimethyloxazolidine-3-carboxylate (17)

The synthetic procedure was analogous to that for **5**; mp 35 °C; *R*_f = 0.75 (*n*-hexane–EtOAc, 3:1).

[α]_D²⁰ +23.5 (*c* 1.5, MeOH).

IR (KBr): 2979 (s), 2936 (m), 2873 (m), 1696 (s), 1477 (m), 1457 (m), 1389 (s), 1365 (s), 1290 (m), 1257 (s), 1173 (s), 1084 (m), 1052 (m), 948 (w), 917 (w), 850 (m), 806 (w), 768 (m), 669 cm^{–1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 3.87–3.79 (m, 3 H, NCH, CH₂), 3.30–3.09 (m, 2 H, CH₂), 2.16–1.89 (m, 2 H, CH₂CH₂I), 1.46–1.21 (m, 15 H, CH₃).¹⁵

¹³C NMR (62.9 MHz, MeOD-*d*₄): δ = 158.1, 101.1, 80.1, 74.1, 70.2, 68.5, 67.9, 67.3, 66.6, 65.3, 64.7, 63.3, 61.1, 56.2, 55.0, 54.4, 53.0, 37.0, 34.4, 33.4, 30.8, 28.8, 27.8, 27.3, 26.4, 26.0, 24.3, 24.1, 24.0, 23.7, 2.0, 0.2, –0.6.¹⁵

Anal. Calcd for C₁₂H₂₂INO₃: C, 40.58; H, 6.24; N, 3.94. Found: C, 40.53; H, 6.23; N, 4.11.

tert-Butyl (4S)-2,2-Dimethyl-4-[2-(pyren-1-yl)ethyl]oxazolidine-3-carboxylate (18)

The synthetic procedure was analogous to that for **7a**; *R*_f = 0.5 (*n*-hexane–EtOAc 5:1).

[α]_D²⁰ +60.6 (*c* 1.0, MeOH).

IR (KBr): 3340 (m), 2976 (m), 2934 (m), 2870 (m), 1693 (s), 1603 (w), 1543 (w), 1509 (w), 1477 (w), 1458 (w), 1387 (s), 1364 (s), 1247 (m), 1206 (m), 1173 (m), 1144 (m), 1095 (m), 1070 (m), 843 (s), 768 (m), 720 (m), 680 (w), 619 cm^{–1} (w).

¹H NMR (250 MHz, DMSO-*d*₆): δ = 8.34–7.94 (m, 9 H, H_{aryl}), 4.10–3.86 (m, 3 H, CHCH₂O, CHCH₂O), 3.45–3.20 (m, 2 H, ArCH₂CH₂), 1.98 (m, 2 H, NCHCH₂CH₂), 1.50–1.40 (m, 9 H, CH₃), 1.18 (s, 6 H, CH₃).¹⁵

¹³C NMR (75.4 MHz, DMSO-*d*₆): δ = 150.9, 135.9, 130.7, 130.2, 129.2, 127.8, 127.3, 127.1, 126.4, 126.0, 124.9, 124.7, 124.2, 124.0, 123.0, 92.6, 79.0, 78.4, 66.1, 59.6, 56.8, 56.4, 35.1, 34.4, 29.1, 27.8, 27.3, 26.6, 24.3, 22.8, 20.6, 13.9.¹⁵

Anal. Calcd for C₂₈H₃₁NO₃: C, 78.29; H, 7.27; N, 3.26. Found: C, 78.08; H, 7.31; N, 3.52.

9H-Fluoren-9-ylmethyl *N*-[(2S)-1-Hydroxy-4-(pyren-1-yl)butan-2-yl]carbamate (19)

The synthetic procedure was analogous to that for **8a**; mp 160 °C; *R*_f = 0.3 (*n*-hexane–EtOAc, 1:1).

[α]_D²⁰ +30.1 (*c* 1.0, DMF).

IR (KBr): 3464 (m), 3304 (s), 3037 (m), 2947 (m), 2870 (m), 1685 (s), 1603 (w), 1544 (s), 1444 (m), 1293 (m), 1268 (m), 1251 (m), 1104 (m), 1084 (m), 1056 (m), 1030 (m), 841 (s), 756 (m), 736 (s), 680 (w), 561 cm^{–1} (w).

¹H NMR (400 MHz, DMSO-*d*₆): δ = 8.36 (d, *J* = 9.6 Hz, 1 H, H_{aryl}), 8.27 (m, 2 H, H_{aryl}), 8.19 (m, 2 H, H_{aryl}), 8.12 (m, 2 H, H_{aryl}), 8.95 (t, *J* = 7.6 Hz, 1 H), 7.95–7.89 (m, 3 H, H_{aryl}), 7.77 (t, *J* = 6.8 Hz, 2 H, H_{aryl}), 7.43–7.31 (m, 5 H, H_{aryl}, NH, D₂O exch.), 4.70 (t, *J* = 5.6 Hz, 1 H, OH, D₂O exch.), 4.42 (m, 1 H, CO₂CH₂CH), 4.30 (m, 2 H, CO₂CH₂CH), 3.62 (m, 1 H, NCH), 3.47–3.35 (m, 3 H, CH₂CH₂-aryl, CH_aH_bOH), 3.24 (m, 1 H, CH_aH_bOH), 2.03 (m, 1 H, ArCH₂CH_a), 1.86 (m, 1 H, ArCH₂CH_b).¹⁵

¹³C NMR (75.4 MHz, DMSO-*d*₆): δ = 156.0, 143.9, 143.8, 140.6, 136.7, 130.8, 130.3, 129.1, 128.8, 127.9, 127.5, 127.3, 127.0, 126.9, 126.3, 125.9, 125.1, 124.8, 124.7, 124.6, 124.1, 123.4, 121.2, 120.0, 65.1, 63.4, 52.7, 46.8, 33.1, 29.2.

Anal. Calcd for C₃₅H₂₉NO₃: C, 82.17; H, 5.71; N, 2.74. Found: C, 82.92; H, 5.71; N, 3.00.

Acknowledgment

Financial support by the Deutsche Forschungsgemeinschaft (SFB 579 'RNA-Ligand Interactions') is gratefully acknowledged. M.S. would like to thank the Dr. Hilmer Foundation for a predoctoral fellowship.

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